

QUANTITATIVE RESEARCH · PRE-REGISTERED · NET OF COSTS

Ghost Whales in Biotech Options

Do large one-sided option trades predict biotech stock moves before the news?

−0.56

In-sample Sharpe, net

−0.76

 Out-of-sample Sharpe,
net (run once)

+1.06%

 Gross 5-day signal;
CI includes 0

23 / 23

 Backtests run
and all disclosed

Summary. We pre-registered a rule that trades biotech stocks in the direction of large one-sided option prints, tested it net of costs in-sample (IS, Jan 2024 – Mar 2026), then ran a locked out-of-sample (OOS) holdout once (Mar – Oct 2026). **The strategy fails after costs in both periods.** The raw signal is positive but not statistically significant, and three things sink it:

- 1. The tail hedge costs more than the edge:** 0.54% vs. 0.40% per trade IS; 1.70% vs. 1.44% OOS (\$5).
- 2. Only long trades earn:** shorts average +0.01% IS and −0.03% OOS (\$5).
- 3. The average rests on five trades:** remove them and it turns negative in both periods (\$5).

Decision: do not deploy. A long-only version is the hypothesis for the next, fresh test.

 1 Why it
should work

2 Data

3 The rule

 4 Results:
it fails

 5 Risk: why
it fails

6 Capacity

7 Next test

1 Economic hypothesis

Thesis. Biotech prices jump on trial readouts, FDA decisions and takeovers. Traders with an information edge may use options for leverage, so large directional option trades can show up *before* the stock moves [1–3].

Who loses. Option market makers, whose delta hedging may push the stock the same way, and stock investors who have not yet absorbed the information [4].

Why it could persist. Option flow is fragmented and costly to read, and news coverage of small-cap biotech is thin, so information diffuses slowly [5].

What is new. We sign each print against the live quote, drop days that news already explains, and price a real options tail hedge.

Registered predictions

- P1** Uncovered whale days earn a positive *net* 5-day return.
- P2** News-covered days earn about zero.
- P3** The edge grows with trade size.
- P4** The edge fades within 10–20 days.

Kill if: net OOS is negative, results flip at nearby parameters, sector beta explains returns, or doubled costs erase it.

2 Data and universe

Source	Used for	Coverage
Massive	Volume screen, news sentiment, option prices for the hedge	1,061 biotech symbols incl. delisted; 599 with options
Databento	Every option trade with the bid/ask at print time (OPRA)	2,452 sampled name-days (\$63.52)
Webull	Split-adjusted daily stock and XBI bars: <i>the only prices traded</i>	Missing 29 signal names (see §3)



Figure 1. Locked split (last 20% of dates). On Oct 3, 2026 the rule was committed (16e75ab), IS results were committed (3c468f6), then the OOS run was executed once.

3 Methodology

TAKEAWAY From 71,294 high-volume option days, 650 trades survive the screen; every input is known before entry.

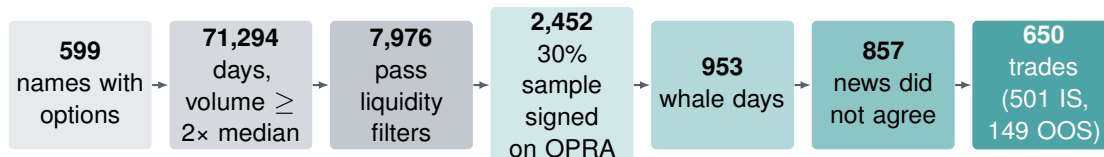


Figure 2. Signal funnel. Of 857 signals, 203 were skipped (102 no price bar, 63 already held, 38 position cap) and 4 could not complete the hold.

1 · Sign each print. Take every option trade with $\geq \$50k$ premium. At or above the ask = *bought*; at or below the bid = *sold*; in between goes to the nearer quote. Bullish premium = calls bought + puts sold.

2 · Whale day. Option volume $\geq 3\times$ normal and $\geq 65\%$ of premium on one side.

3 · News filter. Skip if any article from 3 days before through the next open already agrees with the whales.

4 · Trade. Enter at the **next open**, exit at the open 5 days later. No lookahead.

Survivorship. Delisted names are included. The 86 signals with no Webull bar earned *negative* returns elsewhere, so dropping them flatters us slightly.

Portfolio rules (pre-registered)

Size	1% ÷ daily vol, max 10%; $\leq 2\%$ of ADV; ≤ 10 names
Beta	Net stock dollars hedged daily with XBI
Tail	$\approx 10\%$ OTM, ≈ 30 -day put on longs (call on shorts)
Costs	20 bps/side stock (stress: 40); 2 bps XBI; borrow 5%/yr; hedge at day VWAP + 5%

Overfitting controls. Rules and holdout fixed before any data pull; 16 IS configurations, all reported; 7 OOS runs, all declared in advance; deflated Sharpe ratio [8].

4 Results

TAKEAWAY *The registered strategy loses money in both periods, and doubling costs makes it worse.*

Net of costs	In-sample		OOS	
	base	×2	base	×2
Ann. return	−11.5%	−18.6%	−25.7%	−31.3%
Volatility	18.8%	17.6%	32.3%	30.2%
Sharpe	−0.56	−1.08	−0.76	−1.09
Max DD	−27.8%	−38.9%	−27.3%	−30.1%
Turnover/yr	46×	44×	57×	54×
Trades	501	501	149	149
Hit rate	50.7%	50.7%	51.7%	51.7%

A daily-bar spread estimate [7] suggests ≈ 33 bps half-spread, so the **×2 column is the realistic case**. OOS total return was −15.1% over 139 days. Losses are spread across months rather than one crash (2024: −23.3%; 2025: +0.8%). Turnover counts stock, XBI and option notional over mean equity.



Figure 3. Equity (top) and drawdown (bottom), net of costs, IS then OOS.

Prediction	In-sample	Out-of-sample	Verdict
P1 Signal, 5 days, gross	+1.06% [−0.01, +2.32]	+2.54% [−0.38, +5.98]	NOT SIG.
P2 News days ≈ 0	+3.90%	−3.83%	FAIL
P3 Grows with size (terciles)	+0.57 / +0.24 / +1.20%	−0.15 / +1.35 / +6.82%	MIXED
P4 Fades by 10–20 days	+1.41% at day 20	+3.75% at day 20	FAIL

Table 1. Prediction scorecard (unsized event averages; brackets are 95% bootstrap CIs; $n = 593$ IS, 173 OOS). P1 is positive but not significant, and fails net of costs.

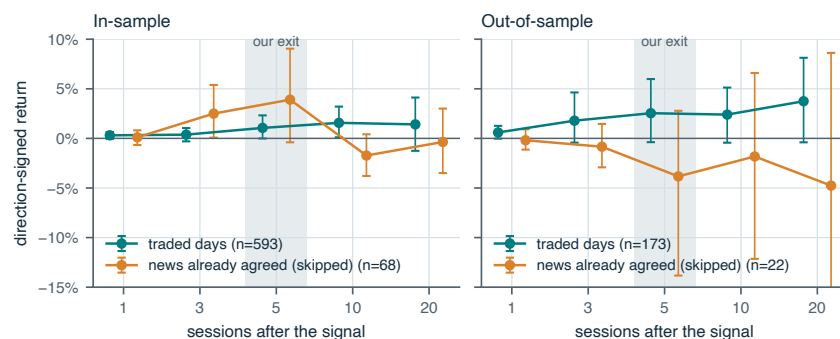
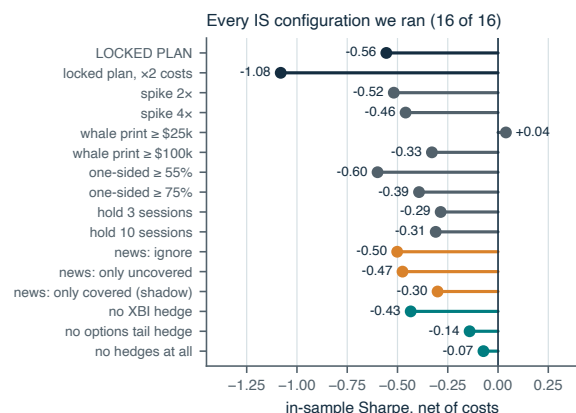


Figure 4. Direction-signed return by horizon (95% CIs). Teal: traded days stay positive and have not faded by day 20 (P4). Orange: news-covered days flip sign between periods (P2).



Robust failure, not bad luck. Every nearby parameter setting also loses IS (left; navy = locked plan). The only one above zero, a \$25k print threshold, has Sharpe +0.04.

Not sector beta. IS XBI beta 0.03 ($t = 0.69$), $R^2 = 0.001$.

No evidence of skill. Deflated Sharpe probability is 0.07 in each period.

5 Risk management

TAKEAWAY *The controls limited tail losses, but the tail hedge cost more than the signal earned.*

Stop	–15% close vs. entry → exit next open. Fired 18× IS, 9× OOS.
De-risk	Halve new sizes when –15% from peak; active 77% of days.
Tail hedge	FDMT lost 67.6% gross but only 20.2% hedged. Yet 27–32% of entries were unhedged (contract did not trade), including the worst OOS loss, BIOA (–54.7%).
Exposure	Median 25–30% of equity in stock; net –80% to +64% before XBI.

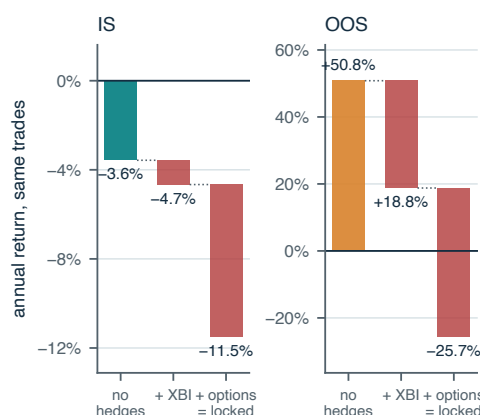
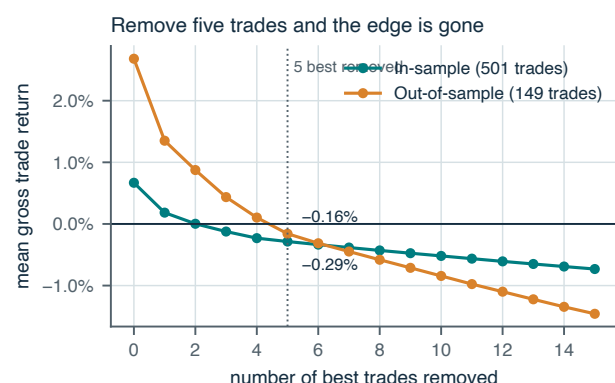


Figure 5. Adding each hedge to the same trades. The tail hedge alone turns OOS from +18.8% to –25.7%/yr.



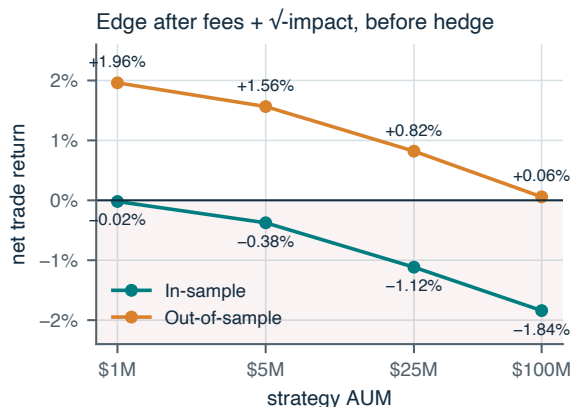
Why it fails.

- Hedge > edge:** net hedge cost 0.54% vs. edge 0.40% per trade IS; 1.70% vs. 1.44% OOS. Median premium 4.4% of position.
- Shorts earn nothing:** +0.01% IS, –0.03% OOS.
- Concentration (left):** the 5 best trades give 142% (IS) and 106% (OOS) of gross P&L.

Controls to add before live trading (*not* backtested): long-only; hedge only when a known catalyst falls inside the hold; cap unhedged names at 5%; stop trading at –20% from peak until a fresh holdout passes.

6 Liquidity and capacity

TAKEAWAY No profitable capacity is demonstrated; option liquidity and borrow bind before stock impact.



AUM	% ADV	Impact (RT)	Capped IS/OOS
\$1M	0.1%	29–32 bps	0 / 0
\$5M	0.5%	64–72 bps	0 / 0
\$25M	2.2–2.5%	138–146 bps	40 / 16
\$100M	8.9–9.9%	211–222 bps	246 / 71

Table 2. Square-root impact model [6], $\sigma_{\text{daily}} \sqrt{\text{shares} / \text{ADV}}$; a scenario, not calibrated fills.

Realised participation was tiny (median 0.10% of ADV, max 2%). Even so, the IS trade is negative at \$1M after impact, before the hedge. The tighter limits are option liquidity (27–32% of chosen hedge contracts never traded) and small-cap biotech borrow, which can far exceed the 5% assumed.

7 Limitations and next steps

Limitations.

- Hedge fills use whole-day VWAP, not known at the open.
- Only 30% of screened days were signed; 139 OOS days, one regime.
- The last IS trade exits Mar 23, overlapping OOS by a few days.
- Quote signing can misread multi-leg trades; news coverage is sparse.

Next test. Register a **long-only, 10-day, catalyst-only-hedge** version and run it on *future* data with no retuning. The hedge-free OOS numbers look good (Sharpe 1.46) but rest on two trades (KOD, AVTX) and were seen after the holdout opened, so we do not adopt them.

BOTTOM LINE. A pre-registered test, net of costs, on a holdout run once: it failed. A fragile long-side signal survives as a hypothesis for new data.

Reproduce. In whale_footprints folder: `pip install -r requirements.txt; set .env; python run_all.py -from-signals (-oos for the locked run); python make_tex_figures.py`. Signal tables are committed, so Databento is not needed.

References

- [1] Easley, D., O'Hara, M. & Srinivas, P. (1998). Option volume and stock prices: evidence on where informed traders trade. *Journal of Finance* 53(2).
- [2] Pan, J. & Poteshman, A. (2006). The information in option volume for future stock prices. *Review of Financial Studies* 19(3).
- [3] Augustin, P., Brenner, M. & Subrahmanyam, M. (2019). Informed options trading prior to takeover announcements: insider trading? *Management Science* 65(12).
- [4] Ni, S., Pearson, N., Poteshman, A. & White, J. (2021). Does option trading have a pervasive impact on underlying stock prices? *Review of Financial Studies* 34(4).
- [5] Hong, H. & Stein, J. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance* 54(6).
- [6] Almgren, R., Thum, C., Hauptmann, E. & Li, H. (2005). Direct estimation of equity market impact. *Risk* 18(7).
- [7] Abdi, F. & Ranaldo, A. (2017). A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies* 30(12).
- [8] Bailey, D. & López de Prado, M. (2014). The deflated Sharpe ratio. *Journal of Portfolio Management* 40(5).

Data and software. Databento OPRA; Massive option, stock and news data; Webull stock/ETF bars. backtrader, pandas, numpy, scipy, matplotlib. Every figure is drawn from committed files in `results/` by `make_tex_figures.py`; no new backtest was run for this version.